

Practice problems #24

1. Let V be a vector space with a basis $B = \{b_1, b_2, \dots, b_n\}$. Show that the coordinate mapping $T : V \rightarrow \mathbf{R}^n$ defined by $T(x) = [x]_B$ satisfies

$$T(cx) = cT(x)$$

for all $x \in V$ and for all scalars c .

2. Find the vector x determined by the given coordinate vector $[x]_B$ and the given basis $B = \{b_1, b_2, b_3\}$.

$$b_1 = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, b_2 = \begin{bmatrix} 5 \\ 2 \\ -2 \end{bmatrix}, b_3 = \begin{bmatrix} 4 \\ -7 \\ 0 \end{bmatrix}, [x]_B = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}.$$

3. Prove that the set $B = \{1 + t^2, t + t^2, 1 + 2t + t^2\}$ is a basis for $\mathbf{P}_2$.

Answers or Hints are on page 2.

#1: Mimic what we did in class.

#2: $\begin{bmatrix} -1 \\ -5 \\ 9 \end{bmatrix}$

#3: Show two things!